Game Modification Task Sheet: Dahdi

Institution: Academic / College Game Development Project
Project: Dahdi – Traditional Indian Mills Board Game
Review Date: September 2026
Evaluator: Faculty / Project Supervisor

Game Reviewed:

Dahdi (Traditional Indian Mills / Alignment Strategy Game)

Game Platform:

Web Browser / HTML5 / CSS3 / JavaScript

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1. Provide a detailed overview of the object of the game.

The fundamental object of Dahdi is to defeat the opposing player through strategic placement and movement of pieces. Each player has nine pieces. Players form mills of three aligned pieces and use successful mills to capture eligible opposing pieces. After placement, the game moves into a movement phase and, when a player has exactly three pieces, a flying phase becomes available. The game ends when the opponent has fewer than three pieces or no legal move.

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2. List 3 or more things you like about the game. Why?

- Clear strategic objective: Mills and captures give every turn a meaningful purpose.
- Simple interaction: The board has a limited number of points, making the controls easy to understand.
- Multiple AI levels: Easy, Medium and Master provide different levels of challenge.
- Local multiplayer: Pass & Play allows two people to use the same device.

3. List things you do not like about the game. Why?

- New-player learning curve: Mills, protected pieces and Flying may be unfamiliar to first-time players.
- No online multiplayer: The current project does not support remote matches.
- Limited personalization: The board has only the implemented visual configuration and optional diagonal variant.
- Limited long-term statistics: The current version does not maintain a full match-history dashboard.

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4. What changes would you make to improve the game?

Modification	**Problem Solved**	**Implementation Detail**
Tutorial / Guided First Match	Reduces confusion for new players.	Add step-by-step prompts for placement, mill formation, capture, movement and Flying.
Online Multiplayer	Adds remote competition.	Add a future server-based multiplayer layer while keeping the existing rules engine reusable.
Enhanced Board Animation	Improves visual feedback.	Add smoother piece movement, capture transitions, mill highlights and phase transitions.

Modification Plan — Continued

Modification	Problem Solved	Implementation Detail
Difficulty Customization	Gives more control over AI challenge.	Add selectable search depth or additional AI behaviour profiles beyond Easy/Medium/Master.
Statistics & Match History	Improves replay value.	Store wins, losses, moves, average match length and selected difficulty in local browser storage.
Theme / Board Skins	Adds personalization.	Add optional traditional visual themes while keeping the current default design.
Guided Hint Explanations	Makes the Hint feature more educational.	Show why a suggested move creates a threat, blocks a mill or improves mobility.

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5. Priority for Future Versions

- Guided tutorial and clearer first-match explanations.
- Improved visual transitions and board animation.
- Optional statistics and match history.
- Additional AI configuration and difficulty controls.

6. Final Modification Assessment

The current Dahdi implementation already contains the essential board-game loop, multiple AI levels, local multiplayer, accessibility controls, localization, audio, undo, hint and rematch functionality. The suggested modifications are therefore focused on improving onboarding, replay value, personalization and future connectivity rather than replacing the core game.

7. Recommended Development Sequence

- First: add a guided tutorial.
- Second: improve board animation and capture feedback.
- Third: add statistics and match history.
- Fourth: add optional visual themes.
- Fifth: evaluate online multiplayer as a separate future module.

Conclusion

The modification plan preserves the existing rules and modular architecture of Dahdi. Each proposed improvement can be added without changing the central board-state and rules engine, allowing the academic version to remain stable while providing a clear path for future development.